σ-Equivariant Elementary Magmas on $AG(2,3)$

Classification, associativity spectrum, and standard elementary selection in a six-rule family

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Abstract

We study a finite elementary family of σ-equivariant row-graded magmas on $M = S \times S$, where $|S| = 3$. The complete space of σ-equivariant cross-row rules has cardinality 3^{18}. The present paper restricts to six distinguished elementary cross-rules $g_1, \ldots, g_6$ and all 27 σ-equivariant diagonal maps. The resulting elementary space

$$\Omega_{\text{elem}} = \{g_1, \ldots, g_6\} \times \Delta_\sigma$$

has 162 objects and decomposes into 90 orbits under diagonal relabelling by S_3.

For this elementary family we derive an explicit master formula for the number of associative triples in terms of the elementary cross-rule and diagonal fixed-point data. Consequently, the associativity spectrum in Ω_{elem} consists of exactly 28 values in $[162, 567]$, all divisible by 3. Among the six standard elementary magmas, two independent criteria—minimum conditional column-access information and minimum associativity—select the PAB magma. A restricted sequential selection theorem further shows that PAB is uniquely selected inside

$$\Omega'_{\text{elem}} = \mathcal{F}_\sigma \times \{g_1, \ldots, g_6\} \times \Delta_\sigma$$

by successively minimizing access information, diagonal entropy, and associativity on the remaining candidate set. Finally, the PAB operation induces an absorption graph on the six off-diagonal states generated by two transitions T_α, T_β, naturally realizing the A_2 root hexagon.

The full 3^{18}-cross-rule universe, full isomorphism rigidity beyond the diagonal S_3-action, operator-level path-memory invariants, and broader Lie-theoretic interpretations remain outside the scope of this paper.

Scope and notation

All classification, orbit-count, master-formula, spectrum, and selection statements below are scoped to one of the following spaces:

$$\Omega_{\text{elem}} = \{g_1, \ldots, g_6\} \times \Delta_\sigma, \qquad |\Omega_{\text{elem}}| = 162,$$

$$\Omega'_{\text{elem}} = \mathcal{F}_\sigma \times \{g_1, \ldots, g_6\} \times \Delta_\sigma, \qquad |\Omega'_{\text{elem}}| = 1458,$$

and the full cross-rule universe

$$\Omega_{\text{full}} = \text{CrossRules}_\sigma \times \Delta_\sigma, \qquad |\,\text{CrossRules}_\sigma\,| = 3^{18}.$$

This paper classifies and analyses Ω_{elem}, not Ω_{full}. The six-rule list is not exhaustive: it is a restricted family of distinguished elementary formulas inside the full set of 3^{18} equivariant cross-row rules.

1 Introduction

1.1 Object of study

Let

$$S = \{0, 1, 2\}, \qquad M = S \times S.$$

Elements of M are written (r, c), with row r and column c. The cyclic shift $\sigma : i \mapsto i + 1 \pmod 3$ acts diagonally on M:

$$\sigma(r, c) = (r + 1, c + 1).$$

We study row-graded binary operations on M, meaning

$$(x \cdot y)_{\text{row}} = x_{\text{row}}.$$

Inside each row we use the three-point Steiner rule, while self-products are controlled by a σ-equivariant diagonal map. Cross-row multiplication is the main classification variable.

A general σ-equivariant cross-row rule has 3^{18} possibilities. We isolate six elementary closed-form rules $g_1, \ldots, g_6$. These six rules are small enough to admit a complete associativity formula, spectrum computation, and selection theory, while still containing the PAB magma. The general background for finite non-associative systems, designs, finite geometries, and root systems is standard; see [Bru58, DK74, BJL99, CD06, Dem68].

1.2 Main results

Theorem (A — PAB structure)**.** *The PAB magma has exactly three idempotents, automorphism group S_3, endomorphism monoid $\{\widehat{\rho} : \rho \in S_3\} \cup \{\varepsilon_0, \varepsilon_1, \varepsilon_2\}$, and associativity count $219/729$. Its non-associativity is row-confined.*

Theorem (B — Elementary classification)**.** *The full set of σ-equivariant cross-row rules has cardinality 3^{18}. Restricting to six distinguished elementary cross-rules and 27 diagonal maps gives $|\Omega_{\text{elem}}| = 162$. Under diagonal relabelling by S_3, this space has 90 orbits.*

Theorem (C — Associativity formula)**.** *For every magma in Ω_{elem}, associativity is determined by the elementary cross-rule and diagonal fixed-point data. The formula is verified over all 162 magmas and all 729 triples.*

Theorem (D — Elementary spectrum)**.** *The associativity spectrum of Ω_{elem} consists of exactly 28 values in $[162, 567]$, all divisible by 3. The extrema 162 and 567 are attained.*

Theorem (E — Standard elementary selection)**.** *Among the six standard elementary magmas, g_1 is the unique rule that simultaneously minimises conditional column-access information and associativity. The selected magma is PAB.*

Theorem (F — Restricted sequential selection)**.** *Inside Ω'_{elem}, successive minimisation of H_{access}, then H_{diag}, and then associativity on the remaining candidate set uniquely selects PAB.*

Theorem (G — Absorption graph)**.** *For $x \in M^{\times}$, the elements absorbed by x are exactly $T_\alpha(x)$ and $T_\beta(x)$.*

1.3 Structure of the paper

Section 2 defines PAB and proves its basic structure. Section 3 constructs the elementary classification space and the diagonal S_3-orbit count. Section 4 derives the row-pattern associativity formula. Section 5 records the elementary associativity spectrum. Section 6 proves the information-theoretic selection results. Section 7 proves the absorption graph theorem. Appendix A gives the computational verification artifacts supporting the finite claims.

2 The PAB magma

2.1 Setup

For each $r \in S$, write

$$F_r = \{(r, 0), (r, 1), (r, 2)\}$$

for the row fiber. For distinct $i, j \in S$, write ij for the unique third element of $S \setminus \{i, j\}$.

2.2 Definition

The PAB magma is the algebra $(M, \cdot)$ defined as follows. Let $x = (r_1, c_1)$, $y = (r_2, c_2)$.

AX0 — row grading.

$$(x \cdot y)_{\text{row}} = r_1.$$

AX1 — cross-row rule. If $r_1 \neq r_2$, then

$$x \cdot y = (r_1, r_2).$$

AX2 — Steiner fiber rule. If $r_1 = r_2$ and $c_1 \neq c_2$, then

$$x \cdot y = (r_1, c_1 c_2).$$

AX3 — diagonal collapse. If $r_1 = r_2$ and $c_1 = c_2 = c$, then

$$(r_1, c) \cdot (r_1, c) = (r_1, r_1).$$

2.3 Idempotents and square fibers

The idempotents are exactly

$$e_r = (r, r), \qquad r \in S.$$

Moreover,

$$F_r = \{x \in M : x^2 = e_r\}.$$

Indeed, $(r, r)^2 = (r, r)$, while $(r, c)^2 = (r, r) \neq (r, c)$ for $c \neq r$.

2.4 Automorphism group

Theorem (2.1)**.** *Every automorphism of the PAB magma is induced by a unique permutation of the three types:*

$$\mathrm{Aut}(M,\cdot) = \{\widehat{\rho} : \rho \in S_3\}, \qquad \widehat{\rho}(r,c) = (\rho(r), \rho(c)).$$

In particular,

$$\mathrm{Aut}(M,\cdot) \cong S_3.$$

Proof. The three idempotents are exactly $e_r = (r,r)$, so any automorphism ϕ permutes them. Write $\phi(e_r) = e_{\rho(r)}$ for a unique $\rho \in S_3$. Since $F_r = \{x \in M : x^2 = e_r\}$, automorphisms preserve square fibers, hence $\phi(F_r) = F_{\rho(r)}$.

For distinct $r, s \in S$,

$$(r,s) = e_r \cdot e_s.$$

Therefore

$$\phi(r,s) = \phi(e_r \cdot e_s) = \phi(e_r) \cdot \phi(e_s) = e_{\rho(r)} \cdot e_{\rho(s)} = (\rho(r), \rho(s)).$$

The same formula holds on diagonal elements because $\phi(r,r) = e_{\rho(r)} = (\rho(r), \rho(r))$. Thus $\phi(r,c) = (\rho(r), \rho(c))$ for all $(r,c) \in M$. Conversely, every such diagonal permutation preserves equality/inequality of rows and columns and the complement relation on S, so it is an automorphism. □

2.5 Endomorphism monoid

Theorem (2.2)**.** *The endomorphism monoid of PAB is*

$$\mathrm{End}(M,\cdot) = \{\widehat{\rho} : \rho \in S_3\} \cup \{\varepsilon_0, \varepsilon_1, \varepsilon_2\},$$

where $\widehat{\rho}(r,c) = (\rho(r), \rho(c))$ *and* $\varepsilon_t(x) = e_t = (t,t)$ *for all* $x \in M$*. Thus* $|\,\mathrm{End}(M,\cdot)| = 9$*.*

Proof. Let $\phi : M \to M$ be an endomorphism. Since endomorphisms preserve idempotents, each $e_r = (r,r)$ is sent to a diagonal idempotent. Define $\rho : S \to S$ by $\phi(e_r) = e_{\rho(r)}$. Because $F_r = \{x : x^2 = e_r\}$, we have $\phi(F_r) \subseteq F_{\rho(r)}$. There are three cases.

If $|\,\mathrm{Im}\,\rho| = 3$, then ρ is a permutation. For $r \neq s$, $(r,s) = e_r \cdot e_s$, so

$$\phi(r,s) = \phi(e_r \cdot e_s) = e_{\rho(r)} \cdot e_{\rho(s)} = (\rho(r), \rho(s)).$$

The same formula holds on the diagonal, hence $\phi = \widehat{\rho}$.

If $|\,\mathrm{Im}\,\rho| = 1$, then $\rho(r) = t$ for all r. For any off-diagonal element $(r,s) = e_r \cdot e_s$,

$$\phi(r,s) = \phi(e_r \cdot e_s) = e_t \cdot e_t = e_t.$$

Also $\phi(e_r) = e_t$ for all r. Hence $\phi = \varepsilon_t$, the constant endomorphism with value $e_t = (t,t)$.

If $|\,\mathrm{Im}\,\rho| = 2$, choose distinct $a, b, c \in S$ with $\rho(a) = \rho(b) = p$, $\rho(c) = q$, and $p \neq q$. In the fiber F_a, the Steiner rule gives

$$e_a \cdot (a,b) = (a,c).$$

Applying ϕ,

$$\phi(e_a) \cdot \phi(a,b) = \phi(a,c).$$

Now $\phi(e_a) = e_p$, and since $(a,b) = e_a \cdot e_b$, we have $\phi(a,b) = e_p \cdot e_p = e_p$. Thus the left-hand side is e_p. On the other hand,

$$\phi(a,c) = \phi(e_a \cdot e_c) = e_p \cdot e_q = (p,q) \neq e_p.$$

This contradiction rules out the two-valued case. The three cases exhaust all maps $S \to S$. □

2.6 Row barrier

Proposition (2.3)**.** *For any $x \in F_r$, $y \in F_s$, with $r \neq s$,*

$$\langle x, y \rangle = F_r \cup F_s.$$

The third row is unreachable.

Proof. By AX0, every product of elements already in $F_r \cup F_s$ remains in one of those two rows. Hence $\langle x, y \rangle \subseteq F_r \cup F_s$.

For the reverse inclusion, first generate F_r. The generator x gives $x^2 = e_r$. If $x \neq e_r$, then x and e_r are two distinct elements of F_r, hence AX2 generates the third element of F_r. If $x = e_r$, then $x \cdot y = (r, s) \neq e_r$, so e_r and (r, s) generate F_r. Thus $F_r \subseteq \langle x, y \rangle$.

The same argument with y gives $F_s \subseteq \langle x, y \rangle$: either $y \neq e_s$ and $y, y^2 = e_s$ generate F_s, or $y = e_s$ and $y \cdot x = (s, r) \neq e_s$ supplies a second element. Therefore $\langle x, y \rangle = F_r \cup F_s$.

□

2.7 Universal identities and row-confinement

Proposition (2.4)**.** *PAB satisfies the following identities for all $x, y \in M$:*

$$(x \cdot y)(x \cdot y) = x \cdot x,$$

$$(x \cdot x)(y \cdot y) = (x \cdot y)(y \cdot x),$$

$$(x \cdot x)(y \cdot x) = (x \cdot y)(y \cdot y).$$

It also satisfies, for all $z, w \in M$,

$$z \cdot (z \cdot w) = (z \cdot w) \cdot z.$$

Proof. Write $e_r = (r, r)$. Since $x^2 = e_{\mathrm{row}(x)}$, the first identity follows from AX0:

$$(x \cdot y)^2 = e_{\mathrm{row}(x \cdot y)} = e_{\mathrm{row}(x)} = x^2.$$

For the second identity, let $r = \mathrm{row}(x)$ and $s = \mathrm{row}(y)$. If $r = s$, then $x \cdot y = y \cdot x$ inside the Steiner fiber, and the second identity reduces to the first. If $r \neq s$, then

$$(x^2)(y^2) = e_r e_s = (r, s),$$

while

$$(xy)(yx) = (r, s)(s, r) = (r, s)$$

by AX1. The third identity is similar. If $r = s$, all factors lie in the same commutative fiber. If $r \neq s$, then

$$(x^2)(yx) = e_r(s, r) = (r, s), \qquad (xy)(y^2) = (r, s)e_s = (r, s).$$

Finally, put $u = z \cdot w$. By AX0, u lies in the same row as z. If $u = z$ the last identity is trivial; if $u \neq z$, it follows from commutativity of the Steiner operation inside that row.

For every triple (x, y, z), both bracketings of the associator stay in the row of x:

$$((x \cdot y) \cdot z)_{\mathrm{row}} = (x \cdot (y \cdot z))_{\mathrm{row}} = \mathrm{row}(x).$$

Thus non-associativity is an intra-row column phenomenon.

□

3 Elementary classification

3.1 Full cross-rule universe

A cross-row input is a quadruple $(r_1, c_1, r_2, c_2) \in S^4$ with $r_1 \neq r_2$. There are 54 such inputs. The diagonal σ-action is free on these inputs, so they form 18 orbits of size 3. A σ-equivariant cross-rule is determined by one output in S on each orbit. Hence

$$|\,\mathrm{CrossRules}_\sigma\,| = 3^{18} = 387,420,489.$$

3.2 Six distinguished elementary cross-rules

This paper restricts to the following six elementary rules:

Rule	Output column	Name
g_1	r_2	column-blind / PAB
g_2	c_2	transparent
g_3	c_1c_2 if $c_1 \neq c_2$, else c_1	complement-transparent
g_4	c_1	echo
g_5	r_1	self-referential
g_6	r_2c_2 if $r_2 \neq c_2$, else r_2	anti-complement

They are selected elementary formulas, not an exhaustive list of equivariant cross-row rules.

3.3 Diagonal maps and Ω_{elem}

A σ-equivariant diagonal map is determined by three base values $(d_0, d_1, d_2) \in S^3$, where $d_k = \delta(0, k)$. Hence $|\Delta_\sigma| = 27$. The standard diagonal is $\delta_{\mathrm{std}} = (0, 0, 0)$, i.e. $\delta(r, c) = r$.

Define

$$\Omega_{\mathrm{elem}} = \{g_1, \ldots, g_6\} \times \Delta_\sigma.$$

Then

$$|\Omega_{\mathrm{elem}}| = 6 \cdot 27 = 162.$$

3.4 Diagonal S_3-orbit count

Theorem (3.1)**.** *Under diagonal relabelling by S_3, Ω_{elem} decomposes into 90 orbits.*

Proof. Burnside's lemma gives

$$|\Omega_{\mathrm{elem}}/S_3| = \frac{1}{6} \sum_{\pi \in S_3} |\,\mathrm{Fix}(\pi)|.$$

The fixed counts are

$$|\,\mathrm{Fix}(\mathrm{id})| = 162, \quad |\,\mathrm{Fix}(\sigma)| = 162, \quad |\,\mathrm{Fix}(\sigma^2)| = 162,$$

and for each transposition τ,

$$|\,\mathrm{Fix}(\tau)| = 18.$$

Therefore

$$|\Omega_{\text{elem}}/S_3| = \frac{162+162+162+18+18+18}{6} = 90.$$

The fixed counts are enumerated in Appendix A.

This theorem concerns the diagonal S_3-action only. It is not asserted to be a classification by arbitrary magma isomorphism.

□

4 Associativity formula in Ω_{elem}

For any $(g, \delta) \in \Omega_{\text{elem}}$, associativity decomposes by row-pattern:

$$\text{Assoc}(g, \delta) = RRR + SRR + RRS + RSR + DIST.$$

The row-pattern counts are:

Pattern	Row condition	Count
RRR	all three rows equal	81
SRR	first row distinct, second and third equal	162
RRS	first and second equal, third distinct	162
RSR	first and third equal, second distinct	162
$DIST$	all rows distinct	162

For δ with base tuple (d_0, d_1, d_2), define

$$p_{\text{dist}} = |\{d_0, d_1, d_2\}|,$$

$$d_s = |\{k \in S : d_k = k\}|,$$

$$\text{fix} = |\{(r, c) : \delta(r, c) = c\}|,$$

and decompose

$$\text{fix} = \text{fix}_{\text{diag}} + \text{fix}_{\text{off}},$$

where fix_{diag} counts fixed pairs (r, c) with $r = c$, and fix_{off} counts fixed pairs with $r \neq c$.

4.1 RRR-formula

The RRR contribution depends only on δ:

p_{dist}	condition on d_s	RRR	fiber associativity
1	any	57	19/27
2	0	33	11/27
2	≥ 1	39	13/27
3	any possible	27	9/27

4.2 Mixed-pattern formulas

Rule	RRS	RSR	SRR	$DIST$
g_1	$9\mathrm{fix}_{\mathrm{off}}$	$9\mathrm{fix}_{\mathrm{off}}$	162	0
g_2	6fix	6fix	6fix	162
g_3	$72-2\mathrm{fix}$	54	54	54
g_4	162	6fix	162	162
g_5	$18\mathrm{fix}_{\mathrm{diag}}$	$18\mathrm{fix}_{\mathrm{diag}}$	162	162
g_6	6fix	$81-3\mathrm{fix}$	6fix	54

4.3 Master formula theorem

Theorem (4.1)**.** *For every $(g,\delta)\in\Omega_{\mathrm{elem}}$, the number of associative triples is determined by g and the diagonal parameters $p_{\mathrm{dist}}, d_s, \mathrm{fix}, \mathrm{fix}_{\mathrm{diag}}, \mathrm{fix}_{\mathrm{off}}$, via the tables above.*

For example,

$$\mathrm{Assoc}(g_1,\delta) = RRR(p_{\mathrm{dist}}, d_s) + 18\,\mathrm{fix}_{\mathrm{off}} + 162.$$

Proof. The five row-patterns partition M^3, so associativity is the sum of the corresponding pattern counts. If a triple has type RRR, both bracketings remain inside one row-fiber; the cross-rule is never invoked. Thus RRR depends only on the one-row operation determined by δ. Since the three row-fibers are σ-translates, it is enough to enumerate the 27 column triples in one fiber. Grouping these triples by equality pattern gives exactly the RRR-table above.

For the mixed patterns $SRR, RRS, RSR, DIST$, AX0 implies that both bracketings always land in the row of the first argument; associativity is therefore a column equality. Substituting each of the six elementary cross-rules reduces the condition to either a constant truth value or to one of the fixed-point counts $\mathrm{fix}, \mathrm{fix}_{\mathrm{diag}}, \mathrm{fix}_{\mathrm{off}}$. This case analysis gives the mixed-pattern table. The computational verification in Appendix A independently enumerates all 162 elementary magmas and all 729 triples for each magma; it reports zero mismatches against these formulas.

□

5 Elementary associativity spectrum

Theorem (5.1 — Elementary associativity spectrum)**.** *Inside Ω_{elem}, the associativity function takes exactly the following values:*

$$\{162, 168, 189, 195, 201, 219, 228, 243, 255, 261, 267, 273, 285, 297,$$
$$309, 351, 357, 363, 381, 459, 471, 489, 513, 519, 531, 543, 561, 567\}.$$

Thus the spectrum has 28 values. Every value is divisible by 3. The extremal values are

$$\min_{\Omega_{\mathrm{elem}}} \mathrm{Assoc} = 162, \qquad \max_{\Omega_{\mathrm{elem}}} \mathrm{Assoc} = 567.$$

Equivalently, after normalisation by $|M|^3 = 729$,

$$\frac{2}{9} \le \frac{\mathrm{Assoc}(g,\delta)}{729} \le \frac{7}{9} \qquad ((g,\delta)\in\Omega_{\mathrm{elem}}).$$

The values divisible by 27 are exactly

$$162, 189, 243, 297, 351, 459, 513, 567.$$

Proof. The master formula of Theorem 4.1 expresses associativity as a finite function of one of the six elementary rules and the 27 diagonal maps. Applying that formula to the 162 pairs $(g, \delta) \in \Omega_{\text{elem}}$ gives the listed set of 28 values. The exhaustive validation in Appendix A independently enumerates all 729 triples for every object of Ω_{elem}, agrees with the formula in every case, and records witnesses for the extremal values. One minimum witness is $(g_6, \delta = (1, 2, 0))$, with associativity 162; one maximum witness is $(g_4, \delta = (0, 1, 2))$, with associativity 567. The divisibility and anchor-value claims are read directly from the verified spectrum table.

These spectrum statements are made only for Ω_{elem}, not for the full 3^{18}-cross-rule universe. □

6 Information-theoretic selection

6.1 Standard elementary selection

For the six standard elementary magmas $(g_i, \delta_{\text{std}})$, define

$$\lambda(g_i) = \frac{\text{Assoc}(g_i, \delta_{\text{std}})}{729}.$$

The conditional access information is

$$H_{\text{access}}(g) = I(C_{\text{out}}; C_1, C_2 \mid R_1, R_2).$$

Rule	H_{storage}	H_{access}	Assoc	λ
g_1	0	0	219	219/729
g_2	$\log_2 3$	$\log_2 3$	273	273/729
g_3	$\log_2 3$	$\log_2 3$	285	285/729
g_4	$\log_2 3$	$\log_2 3$	561	561/729
g_5	0	0	489	489/729
g_6	0	$\log_2 3$	219	219/729

Thus

$$\min H_{\text{access}} = \{g_1, g_5\}, \qquad \min \lambda = \{g_1, g_6\},$$

and

$$\{g_1, g_5\} \cap \{g_1, g_6\} = \{g_1\}.$$

This selects PAB.

6.2 Parameter-free functional uniqueness

For $\beta > 0$, define

$$F_\beta(g) = H_{\text{access}}(g) - \beta(1 - \lambda(g)) = H_{\text{access}}(g) + \beta(\lambda(g) - 1).$$

Corollary (6.1)**.** *Among the six standard elementary magmas, F_β is uniquely minimised by g_1 for every $\beta > 0$.*

Proof. Against g_5, the access costs are equal, while

$$\lambda(g_1) - \lambda(g_5) = \frac{219 - 489}{729} < 0.$$

Therefore $F_\beta(g_1) - F_\beta(g_5) < 0$. For $g \in \{g_2, g_3, g_4, g_6\}$,

$$H_{\text{access}}(g_1) - H_{\text{access}}(g) = -\log_2 3 < 0,$$

and

$$\lambda(g_1) - \lambda(g) \leq 0.$$

Thus

$$F_\beta(g_1) - F_\beta(g) = -\log_2 3 + \beta(\lambda(g_1) - \lambda(g)) \leq -\log_2 3 < 0.$$

So $F_\beta(g_1) < F_\beta(g)$ for every $g \neq g_1$.

□

6.3 Restricted sequential selection in Ω'_{elem}

Let $\mathcal{F}_\sigma$ be the nine σ-equivariant distinct-column within-fiber rules and

$$\Omega'_{\text{elem}} = \mathcal{F}_\sigma \times \{g_1, \ldots, g_6\} \times \Delta_\sigma.$$

Then $|\Omega'_{\text{elem}}| = 1458$.

The verified sequential chain is:

Stage	Criterion	Remaining candidates
0	full Ω'_{elem}	1458
1	minimise H_{access}	486
2	minimise H_{diag} among Stage 1	54
3	minimise associativity among Stage 2	1

The unique final remaining candidate is

$$(g_1, F_{21}, \delta_{\text{std}}) = PAB.$$

Here F_{21} is the complement / Steiner fiber rule.

The statement is sequential: the associativity criterion is applied after the two entropy-based filters. Over all of Ω'_{elem}, the global associativity minimum is 135, not 219, and is not attained by PAB.

7 Absorption graph

Let

$$M^\times = \{(r, c) \in M : r \neq c\}.$$

For $x = (r, c) \in M^\times$, define

$$T_\alpha(r, c) = (c, rc), \qquad T_\beta(r, c) = (c, r).$$

For $x \in M^\times$, define

$$\text{Abs}(x) = \{y \in M^\times \setminus \{x\} : x \cdot y = x\}.$$

Theorem (7.1)**.** *For every $x \in M^\times$,*

$$\mathrm{Abs}(x) = \{T_\alpha(x), T_\beta(x)\}.$$

Proof. Let $x = (r, c) \in M^\times$, so $r \neq c$. If $y = (s, d) \in M^\times$ has $s = r$, then $x \cdot y$ is computed inside F_r. If $y \neq x$, the Steiner rule gives the third element of the fiber, not x. Hence no same-row off-diagonal element is absorbed by x.

If $s \neq r$, then AX1 gives $x \cdot y = (r, s)$. This equals $x = (r, c)$ if and only if $s = c$. The off-diagonal elements with row c are exactly

$$(c, r) = T_\beta(x)$$

and

$$(c, rc) = T_\alpha(x).$$

Therefore $\mathrm{Abs}(x) = \{T_\alpha(x), T_\beta(x)\}$.

□

8 Minimal finite-geometric interpretation

The set $M = S \times S$ is the affine plane $AG(2, 3)$. Each row fiber is a three-point line, and AX2 is the corresponding Steiner completion rule on that line. The six off-diagonal states

$$M^\times = \{(r, c) : r \neq c\}$$

form the usual hexagonal A_2-root pattern. The maps T_α and T_β generate the derived absorption structure on this hexagon.

A Computational verification

A.1 Verification policy

All finite claims in the paper are verified by exact exhaustive enumeration over finite sets. The verification scripts use integer arithmetic and symbolic labels in $S = \{0, 1, 2\}$; no floating-point comparison is used for algebraic claims.

All scripts and generated artifacts are available in the public repository:

`https://github.com/burundai-t/sigma-equivariant-magmas-ag23`

The repository mirrors the directory structure used in this appendix and allows full reproduction of all finite verification results.

Each verification script:

1. defines the finite state space explicitly;
2. enumerates the relevant objects exhaustively;
3. writes stable CSV/log outputs;
4. exits with failure if any required assertion is violated.

A.2 Finite state spaces

Object	Size
$M = S \times S$	9
M^2	81
M^3	729
Cross-row inputs (r_1, c_1, r_2, c_2), $r_1 \neq r_2$	54
σ-orbits of cross-row inputs	18
Full σ-equivariant cross-rules	$3^{18} = 387{,}420{,}489$
Elementary cross-rules	6
σ-equivariant diagonal maps	27
Elementary classification space Ω_{elem}	162
Extended elementary space Ω'_{elem}	1458

A.3 Reproducibility inventory

The artifact root is `proof_artifacts/`. Each block below corresponds to a repository subdirectory; the table lists the script, verified claim, and primary outputs without repeating the full path.

Block	Script	Verified claim	Primary outputs
A1	`cross_rule_count.py`	Full σ-equivariant cross-rule universe has cardinality 3^{18}; the six g_i are distinguished elementary rules.	`cross_rule_orbits.csv` `elementary_rule_vectors.csv`
A2	`a2_burnside_fixed_table.py`	Ω_{elem} has 90 diagonal S_3-orbits under relabelling.	`A2_burnside_fixed_table.csv` `A2_burnside_orbits.csv`

A3	`a3_mixed_pattern_formula_table.py`	Master formula for associativity over Ω_{elem}, decomposed by row-pattern.	`mixed_pattern_formula_table.csv`
			`rrr_formula_table.csv`
			`mixed_pattern_formula_validation.csv`

A4	`a4_spectrum_values.py`	Elementary associativity spectrum has exactly 28 values in the range 162–567.	`spectrum_values.csv`
			`spectrum_witnesses.csv`
			`spectrum_anchor_values.csv`

A5	`omega_prime_selection.py`	Restricted sequential selection in Ω'_{elem}: $1458 \to 486 \to 54 \to 1$.	`omega_prime_fiber_rules.csv`
			`omega_prime_survivors.csv`
			`omega_prime_selection_steps.csv`
			`omega_prime_step3_winner.csv`
			`omega_prime_global_min_assoc.csv`

S1	`p1_polishing_verification.py`	Automorphism rigidity, endomorphism monoid, standard elementary selection, symbolic F_β, and absorption graph checks.	`automorphism_table.csv`
			`endomorphism_table.csv`
			`standard_canonical_selection.csv`
			`fbeta_symbolic_comparisons.csv`
			`absorption_graph_table.csv`

A.4 Corrected cross-rule count

The cross-row input set is

$$\Sigma = \{(r_1, c_1, r_2, c_2) \in S^4 : r_1 \neq r_2\}, \qquad |\Sigma| = 54.$$

The diagonal cyclic action of σ on Σ is free, so it has 18 orbits. A σ-equivariant cross-rule is determined by one value in S on each orbit, hence

$$|\mathrm{CrossRules}_\sigma| = 3^{18} = 387{,}420{,}489.$$

The corresponding verification block is `A1_cross_rule_count`. Thus the six rules $g_1, \ldots, g_6$ are distinguished elementary cross-rules, not the full set of σ-equivariant cross-rules.

A.5 Burnside count for Ω_{elem}

For

$$\Omega_{\mathrm{elem}} = \{g_1, \ldots, g_6\} \times \Delta_\sigma, \qquad |\Omega_{\mathrm{elem}}| = 162,$$

the diagonal S_3-relabelling fixed counts are:

Group element	Fixed objects
identity	162
3-cycle σ	162
3-cycle σ^2	162
transposition τ_{01}	18
transposition τ_{02}	18
transposition τ_{12}	18

Thus Burnside gives

$$\frac{162 + 162 + 162 + 18 + 18 + 18}{6} = 90.$$

The verified statement is that Ω_{elem} decomposes into 90 diagonal S_3-orbits. These are not asserted to be full magma isomorphism classes.

A.6 Associativity master formula

For each $(g, \delta) \in \Omega_{\mathrm{elem}}$, the computation enumerates all 729 triples and decomposes associativity into row-patterns

$$RRR, \quad SRR, \quad RRS, \quad RSR, \quad DIST.$$

The formula table is validated for all 162 objects. The validation output includes

```
Formula mismatches: 0
```

The corresponding verification block is `A3_mixed_pattern_formula`.

A.7 Associativity spectrum in Ω_{elem}

The verified spectrum is

$$162, 168, 189, 195, 201, 219, 228, 243, 255, 261, 267, 273, 285, 297,$$
$$309, 351, 357, 363, 381, 459, 471, 489, 513, 519, 531, 543, 561, 567.$$

It contains exactly 28 values. The minimum is 162, the maximum is 567, and all values are divisible by 3. The values divisible by 27 are

$$162, 189, 243, 297, 351, 459, 513, 567.$$

The corresponding verification block is `A4_spectrum`.

A.8 Standard elementary selection

The standard elementary selection table is:

Rule	H_{storage}	H_{access}	Assoc at δ_{std}	Selection role
g_1	0	0	219	selected
g_2	$\log_2 3$	$\log_2 3$	273	rejected
g_3	$\log_2 3$	$\log_2 3$	285	rejected
g_4	$\log_2 3$	$\log_2 3$	561	rejected
g_5	0	0	489	rejected by associativity
g_6	0	$\log_2 3$	219	rejected by access cost

Thus

$$\min H_{\text{access}} = \{g_1, g_5\}, \qquad \min \lambda = \{g_1, g_6\},$$

and the intersection is $\{g_1\}$. The normalized table and the symbolic F_β-comparisons are recorded in the structural verification outputs listed above.

A.9 Restricted sequential selection in Ω'_{elem}

The extended elementary space is

$$\Omega'_{\text{elem}} = \mathcal{F}_\sigma \times \{g_1, \ldots, g_6\} \times \Delta_\sigma, \qquad |\Omega'_{\text{elem}}| = 1458.$$

The nine fiber rules are parametrized by seeds

$$u = f(0,1), \qquad v = f(0,2),$$

with equivariance determining all other distinct-column products. The PAB/Steiner rule is F_{21}.

The verified sequential chain is:

Stage	Criterion	Remaining candidates
0	full Ω'_{elem}	1458
1	minimise H_{access}	486
2	minimise H_{diag} among Stage 1	54
3	minimise associativity among Stage 2	1

The unique final remaining candidate is

$$(g_1, F_{21}, \delta_{\text{std}}) = \text{PAB}.$$

The verification also records that the global associativity minimum over all Ω'_{elem} is 135 and is not attained by PAB. Therefore the selection theorem is stated sequentially.

A.10 Additional structural checks for PAB

The structural verification block checks the finite parts of the PAB-specific claims used in the main text. It enumerates the six diagonal automorphisms, the six automorphisms plus three constant endomorphisms, the standard elementary selection table, the symbolic F_β-comparisons, and the six-row absorption graph table.

B Open problems and further directions

1. Determine whether the 90 diagonal S_3-orbits in Ω_{elem} coincide with full magma isomorphism classes. Equivalently, prove or refute the required rigidity lemma.
2. Classify or stratify the full 3^{18}-element σ-equivariant cross-rule universe.
3. Determine whether the free PAB-magma on two generators is finite or infinite.
4. Define and analyze operator-level path-memory invariants for PAB and for restricted elementary families.
5. Develop the operator linearization theory of PAB, including left-multiplication operators and intertwining spaces.

Note on AI assistance

This work was developed with the assistance of large language models.

The author specified the conceptual framework, constraints, and target properties of the objects under study. Within this framework, AI tools were used for computational exploration, code drafting, and structuring intermediate results.

All mathematical definitions, statements, and theorems presented in the paper were established and independently verified by the author, with exhaustive finite checks (Appendix A).

References

[BJL99] Thomas Beth, Dieter Jungnickel, and Hanfried Lenz. *Design Theory*. Cambridge University Press, 2 edition, 1999.

[Bru58] R. H. Bruck. *A Survey of Binary Systems*. Springer, 1958.

[CD06] Charles J. Colbourn and Jeffrey H. Dinitz, editors. *Handbook of Combinatorial Designs*. Chapman & Hall/CRC, 2 edition, 2006.

[Dem68] Peter Dembowski. *Finite Geometries*. Springer, 1968.

[DK74] J. Dénes and A. D. Keedwell. *Latin Squares and Their Applications*. Academic Press, 1974.